

Your grade: 92%

Your latest: 80% • Your highest: 92% • To pass you need at least 80%. We keep your highest score.

Next item →

coach

Almost there! Let's review your answers and then try again.

Your work showed skill in understanding transfer learning, tensor manipulation, and optimization techniques, but there is room for growth in model performance improvement techniques. Before retrying this assignment, we recommend focusing on this key area:

- Layer Calculations (Coding):** In Question 3, your understanding of the PyTorch functions commonly used to improve model performance was incomplete. Try reviewing [this item](#) to refresh your memory.

Good job.

Retry assignment



1. Which of the following best defines transfer learning in the context of neural networks?

1 / 1 point

- ☐ Combining multiple models to handle different parts of a task.
- ☒ The process of adapting a pretrained model to perform a new but similar task.
- ☐ Using unsupervised learning to improve model performance.
- ☐ Training a model from scratch using a large dataset.

✔ Correct

Correct! Transfer learning involves adapting a pretrained model to a new, yet similar task.

2. Which method is used to freeze the layers of a pretrained model in PyTorch?

1 / 1 point

- ☐ Changing the activation functions to linear.
- ☐ Removing the layers from the model's architecture.
- ☐ Setting the `train` attribute of the layers to `False`.
- ☒ Setting the `requires_grad` attribute of the parameters to `False`.

✔ Correct

Correct! Freezing layers involves setting the `requires_grad` attribute of the model's parameters to `False`.

3. Which of the following PyTorch functions are commonly used to improve model performance?

0 / 1 point

- ☐ `unsqueeze`
- ☐ `torch.no_grad`
- ☐ `view`
- ☒ `max_pool2d`

✘ This should not be selected

Incorrect. The `max_pool2d` function is typically used in convolutional neural networks (CNNs) for down-sampling, not specifically for improving model performance in LSTMs.

- ☒ `batch_norm`

✘ This should not be selected

Incorrect. While `batch_norm` is useful in many contexts, it is not typically associated with improving performance in LSTM models.

4. Which PyTorch function is used to change the shape of a tensor without changing its data?

1 / 1 point

- ☐ `reshape`
- ☒ `view`
- ☐ `squeeze`
- ☐ `expand`

✔ Correct

Correct! The `view` function in PyTorch is used to reshape a tensor without altering its data.

5. What is the primary purpose of using the Adam optimizer in training an LSTM model?

1 / 1 point

- ☐ To maintain a constant learning rate
- ☒ To adapt the learning rate for each parameter
- ☐ To increase the model's complexity
- ☐ To reduce the number of training epochs required

✔ Correct

Correct! The Adam optimizer adjusts the learning rate for each parameter, which helps in faster and more efficient convergence.